Artificial Intelligence Lab (AI-2002)

Lab Task 10

TASK 1: Student Performance Prediction

You are working as a data scientist for a university. Your goal is to predict a student's final exam score (continuous variable) based on various factors such as study hours, attendance, previous grades, participation in class, and internet usage.

Perform the following tasks:

- Clean the dataset and handle missing values (e.g., missing attendance or study hours).
- Encode categorical variables (e.g., participation level: Low, Medium, High).
- Identify the most important features affecting student performance.
- Train a regression model to predict the final score.
- Evaluate the model using appropriate metrics (e.g., MAE, RMSE, R^2).
- Predict the final score for a new student given their features.

TASK 2: Loan Approval Classification

You are working for a bank to automate loan approvals. The dataset contains customer details such as income, employment status, credit score, loan amount, and marital status. The goal is to classify whether a loan should be approved (1) or rejected (0).

Perform the following tasks:

- Preprocess the dataset:
    - Handle missing values
    - Encode categorical variables (e.g., employment status, marital status)
- Perform feature selection to identify key factors influencing loan approval
- Train a classification model (e.g., Logistic Regression, Decision Tree)
- Split the dataset into training and testing sets
- Evaluate the model using metrics like accuracy, precision, recall, and F1-score
- Use the model to predict loan approval for a new applicant

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TASK 3: Customer Churn Prediction

You are working for a telecom company. Your task is to predict whether a customer will churn (leave the service) or not.

The dataset includes:

- Monthly charges
- Contract type
- Tenure (how long the customer has stayed)
- Internet service type
- Customer support calls

Perform the following tasks:

- Clean the dataset:
  - Handle missing values
  - Detect and treat outliers
- Convert categorical variables into numerical form
- Perform feature scaling if needed
- Split the dataset into training and testing sets
- Train a classification model (e.g., SVM, Random Forest)
- Find a decision boundary (hyperplane if applicable)
- Extract rules or feature importance from the model
- Evaluate performance using confusion matrix, accuracy, precision, recall, and F1-score
- Predict whether a new customer will churn